

PhD Stage 2 Proposal

Computational Modelling of Multilateral Treaties

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Proposed thesis title and type

“Computational Modelling of Multilateral Treaties”, thesis by publication.

Proposed supervisory team and their credentials

Role	Name	Affiliation
Principal Supervisor	Prof. Michael Bode	School of Mathematical Sciences
Associate Supervisor	Prof. Jonathan Rhodes	School of Biology and Environmental Science

Principal Supervisor — Prof. Michael Bode Prof. Bode is Deputy Director of Capability Development at SAEF (Securing Antarctica’s Environmental Future). His work centres on applied mathematics and conservation, with a focus on decision-making under uncertainty and mathematical biology. His previous work on Antarctic Treaty System (ATs) decision-making and conservation makes him an expert on consensus-based systems.

Associate Supervisor — Prof. Jonathan Rhodes Prof. Rhodes is Director of the Centre for Environment and Society. He specialises in spatial ecology, environmental governance, and the modelling of coupled human–natural systems, bridging quantitative models with real-world policy. His expertise gives him insight into the policy implications of environmental decisions, grounding the treaty-system applications of this research.

Chapter 1

Background and Literature Review

1.1 Introduction

Consensus is a collective decision-making process where every party can object to any proposal and effectively veto it. International organisations are the standing forums through which states take collective decisions. One becomes a consensus regime when consensus is the only means to reach them. Importantly, consensus is not the same as unanimity: a unanimity rule puts a proposal to a vote and records who votes in favour, against or abstain, whereas under consensus a decision proceeds whenever no party registers a formal objection (Rietig et al. 2023). Since a single party can halt any proposal, an outside observer might expect consensus regimes to decide very little. In practice, Sykora-Bodie and Morrison (2019) show that consensus regimes can reach agreement even on contested, transnational issues, relying on informal practices (trust, dialogue, leadership) that fill the gaps left by their formal rules. But success is never guaranteed and even historically productive regimes can stall (Gardiner et al. 2025). Quantitatively, the picture is no different: what separates a consensus regime that keeps reaching decisions from one that can no longer reach them is not clearly explained by existing models of consensus dynamics (Pérez et al. 2018; Tran et al. 2024). A model that better reproduces how consensus is reached or broken is therefore needed, both to explain past behaviour and to anticipate when a regime approaches the limits of its capacity to act.

Models of multi-actor decision-making already exist, but they were not built to explain why a consensus regime stops reaching decisions: they represent each actor with a single, low-dimensional vector (DeGroot 1974; Tsebelis 2002) and rarely use the actual text those actors produce (Botelho et al. 2026; Bernardo et al. 2021), which is where they state what they want. To reproduce how a consensus regime behaves, a model needs positions across many issues, read directly from the text the actors produce, and

validated against observed outputs rather than assumed. Recent advances in natural language processing and machine learning (Devlin et al. 2019; Grootendorst 2022) make each of these upgrades feasible.

The Antarctic Treaty system (ATs) is a consensus-driven regime. At its centre is the Antarctic Treaty (AT), whose consultative parties meet yearly at the Antarctic Treaty Consultative Meetings (ATCM), where outputs are adopted only by consensus (Secretariat of the Antarctic Treaty 2023). Historically, the ATCM has been productive in terms of decisions reached. However, in 2022, for the first time, the parties failed to reach consensus on the final report, and even as activity has increased, the regime has become less responsive to challenges (Gardiner et al. 2025). The ATs is therefore a consensus regime approaching the limits of its capacity to act, and it is the case study of this thesis.

1.2 Modelling consensus: responsive and anticipatory approaches

There are two traditional groups of models for how groups reach consensus. The first group is responsive and interactive: each actor follows an update rule with no objective at all: it repeatedly reacts to the current positions of the others (each position being that actor’s opinion expressed as a number). These are the opinion-dynamics models. They treat consensus as *convergence*: opinions are iteratively averaged until they meet (DeGroot 1974), with several variants (Appendix E). Opinion-dynamics models can nevertheless test for agreement: declare consensus when all final positions lie within a tolerance radius, as Bernardo et al. (2021) do for the negotiation of the 2015 Paris Agreement. That test, however, captures numerical convergence. It does not model a proposal that is entirely unacceptable to a party. In the ATs, any party can block a decision no matter how close the remaining positions are.

The second group of models is anticipatory: each actor reasons about how the others will respond and chooses its actions to maximise an objective function. These models use game theory to treat consensus as a unanimity rule: in the veto-players framework of Tsebelis (2002), the status quo is replaced only when no party is left worse off (formal condition in Appendix E). Each party accepts only the policies it prefers to the status quo, so the set of adoptable policies is the intersection of these per-party sets. As parties multiply and their positions spread, the sets overlap less and the intersection shrinks (Tsebelis 2002).

Each group of models has strengths and limits. Responsive models are usually low-dimensional, can be scaled to many agents, and are easy to calibrate against data. But

their agents have no objective and no strategy. Anticipatory models represent vetoes and strategic choice directly, and they explain *why* a proposal passes or fails. But their equilibria become hard to compute as parties and issues multiply, and they are rarely confronted with data from real negotiations.

The ATs is this thesis’s case study of such a regime: its Consultative Parties (CPs) take all decisions by consensus at the ATCM.

Between these two groups lies a middle ground: models that retain the strategic, veto-bearing structure of the second group but learn equilibrium from repeated interaction instead of solving for it. This field is treated in the next section.

1.3 Negotiation and coalition formation simulation using reinforcement learning

The limits set out in Section 1.2 imply two requirements for a model of ATs behaviour. First, the model must represent many heterogeneous parties, each holding a veto and bargaining over a status quo (i.e. if no proposal passes, the current rules stay in force, so each party judges a proposal against them). Second, it must also remain computable with that many parties and issues. Those two requirements cannot be tackled with the closed-form equilibria of analytic game theory. Multi-agent reinforcement learning (MARL) meets both requirements: it keeps the game-theoretic substance but learns equilibrium behaviour from repeated interaction instead of solving for it.

Formally, MARL is defined over stochastic (Markov) games, a generalisation of repeated games in which the state of the negotiation changes as the agents act. Each agent observes the state, chooses an action, and receives a reward: the objective function it learns to maximise. The learning procedures approximate equilibrium behaviour over many simulated rounds rather than deriving it in closed form (Littman 1994).

The MARL literature spans foundational bargaining and coalition-formation work (Gronauer and Diepold 2022; Sarkar et al. 2022) and, most recently, Kulkarni et al. (2025) uses agents based on Large Language Models (LLMs) to model political negotiation directly (see Appendix E for more applications). However, MARL simulations run on synthetic data or abstract benchmarks¹ (Bolleddu 2025) rather than the deliberative text records of a real consensus regime.

¹Standard test environments such as the board game Diplomacy.

1.4 Quantitative and text-based study of institutions and treaties

The quantitative study of international institutions long predates its application to Antarctica. One line of work codes the design of treaties and organisations, such as their decision rules, as structured variables, and links these variables to institutional performance, survival, and decline (Gray 2018). A more recent trend treats the *text* of international agreements and debates as data (Grimmer and Stewart 2013), recovering state positions from voting records and speeches (Baturo et al. 2017; Slapin and Proksch 2008). Both suggest that the preferences and design features of treaty regimes can be measured, not merely described.

The ATs’s health has been assessed qualitatively (Mancilla and Jabour 2023) and quantitatively (Gardiner et al. 2025; Botelho et al. 2026; Kumar et al. 2026), in two ways. Gardiner et al. (2025) measures the regime in aggregate, counting inputs, outputs, and the time legally binding Measures take to enter into force, and concludes that responsiveness to challenges in the Antarctic region is declining. The second way is structural. A Working Paper (WP), the document type through which parties bring proposals to the ATCM (Appendix F), may be submitted by a single party or co-authored by several, and this co-authorship has been used to map which parties collaborate with which through co-authorship networks of ATCM documents (Botelho et al. 2026; Kumar et al. 2026). This thesis uses co-authorship methods as validation targets: parties with similar stances (directional positions on a topic) are likely to co-author papers together, and the same measures should help predict the output series reported by Gardiner et al. (2025) (Section 2.1.2). Neither analysis, however, reads the content of the papers, so neither can say which issues divide the parties, a limitation taken up in Section 1.5.

A separate field does read the content: positional measures place each actor on a policy space using only the text the actor produces. It has two paradigms, which differ in how they turn text into numbers: word counts (i.e. frequencies) or embeddings, vectors in which texts with similar meaning end up close together. By contrast, the word-count paradigm recovers a single, uninterpreted dimension, which in a corpus as topically diverse as the ATCM Working Papers would likely reflect what parties talk about rather than what they support. This thesis adopts the embedding paradigm: topics are discovered with a pipeline (Grootendorst 2022), and stances are read with natural language inference (NLI) classifiers, which judge whether a text supports or contradicts a stated hypothesis (Burnham 2023; Abercrombie and Batista-Navarro 2022). The result is a topic-conditioned stance: a separate position per party and topic through time. The supporting literature and definitions are reviewed in Appendix G,

and Section 2.1.2 sets out the validation checks.

1.5 Summary of literature gap

The literatures reviewed above differ in method, but none recovers a directional preference: which country is for or against what, extracted from the regime’s own words (their limitations are collected in Appendix E).

This gap has two parts. The knowledge gap is the question raised in Section 1.1: whether the regime’s outputs, including its failures, can be anticipated from CPs’ positions at all. The methodological gap is the missing instrument: no existing method measures each party’s directional stance, per topic and over time, from a consensus regime’s own textual records, the ATs included. Without that instrument the question cannot be answered. Section 1.6 turns this gap into three research goals.

The closest prior work shares parts of this design but not its object. Stance detection recovers for-or-against positions from parliamentary debates (Abercrombie and Batista-Navarro 2022), but not a per-party, per-topic measure through time. Moghimifar et al. (2024) drive a negotiation simulation with preferences derived from party manifestos, but in a parliamentary coalition setting, not a consensus regime where every party holds a veto. No prior work extracts a directional preference for each party over time from a consensus regime’s own textual records, nor uses such preferences to drive a strategic simulation in which agents have veto power. This thesis supplies both halves and joins them: the preferences recovered from the record become the parameters of the simulation that reproduces consensus and its failures.

1.6 Research problem

The first goal of this thesis is to reconstruct parties’ preferences, so the regime’s decisions and failures can be anticipated from them. These preferences are extracted from the written records as a directional measure (stance) over issues, understood as topic–axis pairings, and across time. The second goal is to extract directional opposition (veto) signals from the same records, recovering non-adoption and deferral as observables, and to test whether the support and opposition signals together reproduce documented episodes. The third goal is to assemble these measures into a decision-support tool that, for an unseen proposal, ranks the parties most likely to object and the axis on which the objection is likely to fall, evaluated prospectively against held-out proposals. The fourth goal is to use these preferences to parameterise a multi-agent model that simulates consensus and its failures.

Chapter 2

Program and design of the research project

2.1 Objectives, methodology and research plan

The countries that participate in the Antarctic Treaty Consultative Meetings (ATCMs), and that may submit Working Papers (WPs) for discussion at these meetings, are known as Consultative Parties (CPs). A WP is, in principle, a proposal in textual form. Only the CPs can block a proposal from reaching consensus, by objecting to it, and a single objection is enough to prevent adoption. For further details on how the ATCMs' consensus-based system operates, see [Appendix F](#).

Consensus regimes can stall, as the ATCM does. For example, the designation of the emperor penguin as a protected species has been debated since 2022, and WPs have been submitted since then, but no consensus has been reached (United Kingdom et al. 2024; Antarctic Treaty Consultative Meeting 2024; Antarctic Treaty Consultative Meeting 2025). Over a comparable period, the emperor penguin population in Antarctica declined by an estimated 9.6% between 2009 and 2018 (LaRue et al. 2024). A stalled regime is therefore a problem on substantive matters.

A party seeking to get a proposal adopted must anticipate every CP's response to the WP. This makes it important to propose text that is likely to be acceptable to all CPs. In these consensus regimes there are no voting records from which to derive the countries' positions (Secretariat of the Antarctic Treaty 2023). Moreover, a Working Paper serves to open a discussion, but who leads that discussion, and what is said within the ATCM's working groups, is not recorded. What is available is the paper trail: the Working Paper, the final report, and the Measures, Resolutions and Decisions that follow (Antarctic Treaty Consultative Meeting 1995), all publicly available through

the ATS website.

However, this database contains thousands of documents (Gardiner et al. 2024), so reading them manually is a real challenge. Manual reading also does not support a systematic analysis of the past decades of ATCM deliberation. It is therefore difficult for external observers and for CPs to establish who sets the agenda at the ATCM, how discussions are typically settled, and how parties bargain with one another. Admittedly, countries hold their own internal records, but these can be missing or biased, and recently accepted CPs do not hold the full record that longstanding CPs do. A new approach that processes the text and is reproducible is therefore needed.

Natural language processing (NLP) has emerged as a possible solution to this problem. The approach has gained momentum in several international relations applications, where text is analysed to extract parties' positions (Grimmer and Stewart 2013; Slapin and Proksch 2008; Baturio et al. 2017; Rheault and Cochrane 2019). It allows a quantitative measure that is reproducible and feasible for other corpora in political science and international relations.

This research project develops a decision-support tool for parties proposing changes to the current settlement of particular matters in consensus-based international forums. The tool is constructed from, and evaluated against, the documentary record of the ATCMs, which serves as the main case study.

For this project I define:

- **Topic:** a matter of collective attention among Consultative Parties. It is a noun, e.g. 'mineral'.
- **Axis:** the action on a *topic*. For example, 'extraction' or 'regulation'.
- **Issue:** a *topic* joined to an *axis*: a matter of attention that has become a question of contention. For example, 'mineral extraction'.
- **Stance:** a scalar value representing the directional attitude towards an *issue*: a number on a line running from 'for' (+1) to 'against'(-1). For example, the USSR's *stance* on mineral extraction is +0.5, that is, for.
- **Status quo:** the current institutional settlement of an *issue*. For example, at the time a Party proposed to regulate mineral extraction, any company or country wishing to extract minerals in Antarctica had no framework or law to operate under. The *status quo* for that *issue*, at that time, was therefore that extraction was unregulated.
- **Proposal:** a text that puts forward a change to the *status quo*. For example, a Working Paper proposing the regulation of mineral extraction.
- **Blocking statement:** a passage in a final report recording an objection by a CP, or a group of CPs, named or unnamed, that prevented a *proposal* from being

adopted at that meeting. For example: “Many Consultative Parties stated that they did not consider Belarus to have fulfilled the requirements outlined in Article IX (2) of the Antarctic Treaty and in Decision 2 (2017)” (ATCM Final Report 47, 2025).

2.1.1 Chapter 1: Estimating Parties’ Stance

Motivation Under the ATCM Rules of Procedure (Secretariat of the Antarctic Treaty 2023), the principal formal mechanism for changing the *status quo* is the submission of a Working Paper. These *proposals* are likely to contain their authors’ views on how an *issue* should be settled: a WP names the *issue* and expresses its authors’ *stance* on it. Recovering those *stances* offers one way to reveal patterns in the parties’ positions, and in what they treat as relevant and to what extent.

The recovered *issues* and *stances* would allow a proposing party to identify possible allies in its efforts to change the *status quo*. They would also provide the quantitative basis on which more sophisticated tools can be built (see Chapter 2 and Chapter 3).

Substantive aim To describe each ATCM party’s *stances* on particular *issues* relative to the other parties, and to compare how these change over time.

Methodological objective To extract the authors’ *stances* from the Working Papers submitted by the parties, using Natural Language Processing (NLP) techniques.

Assumption The CPs express their *stances* on particular *issues* in the papers they submit.

Research questions

- a. Can an NLP pipeline recover the parties’ *stances* on particular *issues*?
 - i. Which *issues* can be extracted from the CPs’ Working Papers across six decades of ATCM deliberation?
 - ii. Can a *stance* per CP on each *issue* be recovered from the WPs?
- b. Can parties be grouped by the similarity of their *stances* on a given *issue*?

Approach The approach to answering RQ (a) is to build a pipeline with multiple stages (full details and notation in [Appendix E](#)). The first stages of the pipeline discover what the corpus is about: each document is embedded into the vector space introduced in Section 1.4 (detailed in [Appendix G](#)). Then BERTopic (Grootendorst 2022) clusters the embedded corpus into a set of *topics*. Each *topic* has a centroid: the average embedding of its documents.

The next stages derive the *issues* from the *topics*, then select the passages relevant to each *issue*, gated by cosine relevance to the *issue* centroid. The *issue*-relevant sentences of each paper are scored in one pass by a Natural Language Inference (NLI) classifier against two author-anchored hypotheses per *issue* (that the author supports it, and that the author opposes it). It outputs a distribution over three ordinal classes: “For”, “Neutral” or “Against”, whose expected value is the *stance* (see [Appendix E](#) for more details).

RQ (b) is answered by grouping similar *stances* per *issue*. Grouping by *stance* value provides a set of CPs that held a similar *stance* at a given ATCM. The grouping is checked against observed co-authorship: parties with similar *stances* on an *issue* should be the ones that co-author papers on it.

Preliminary work on this chapter is complete. [Appendix H](#) shows the results.

2.1.2 Chapter 2: Estimating Parties’ Blocking Behaviour and Validating Against Documented Episodes

Motivation When submitting a *proposal*, a CP has no clear view of the objections other parties are likely to raise, or of which parties will raise them. Experts and national delegations may hold institutional memory and internal records that offer some indication, but these can be mistaken, partial or lost. On the basis of the textual record alone, the only way to form such a view at present is to read through thousands of Working Papers and final reports. Beyond being impractical, this supports no quantitative analysis of how positions have shifted over time, and offers no basis for anticipating how a new *proposal* will be received.

At various points in the history of the ATCMs, contested *issues* have been proposed and subsequently adopted or blocked. Several of these episodes have attracted the attention of external observers and scholars, who have documented them in detail. Here, “episode” borrows its sense from computer science: a single sequence of events, where interactions between agents and their environment occur, that takes the system from an initial state to a final one. Some ATCM episodes have unfolded across several successive meetings. Such episodes show the consensus-based system at its most characteristic: a *proposal*

is put forward, discussed, rejected, and resubmitted at a later ATCM, and the cycle repeats until the *issue* is settled. Because these episodes are documented independently of the ATCM record itself, they provide a point of validation for analyses derived from that record.

Substantive aim To establish how likely a proposed change to the *status quo* is to be blocked by specific parties.

Methodological objective First, to identify *blocking statements* in the ATCM final reports, together with the parties that made them. Second, to estimate the probability that a party *c* will block a *proposal d* on *issue k*. Third, to validate the *stance* and blocking probability measures against documented episodes.

Assumption The final reports record *blocking statements* in sufficient number and attributability to build a predictive model with non-trivial accuracy.

Research questions

- a. How can *blocking statements*, and the parties that made them, be identified in ATCM final reports?
 - i. How reliably can *blocking statements* be traced back to the *proposals* that prompted them?
 - ii. What proportion of *blocking statements* in final reports can be attributed to an identifiable party?
- b. Can a blocking probability per party and *issue* be estimated from the *blocking statements*?
- c. How well do the *stance* and blocking probability measures reproduce the direction of party positions and the composition of blocs across meetings, when compared against a pre-specified set of documented episodes?

2.1.3 Chapter 3: Simulating Consensus: Coalition Formation and Decision Support

Motivation A decision-maker in a consensus-based international forum constantly faces the challenge of proposing changes that are minimally acceptable to every other party. Meeting it requires knowing what other parties have proposed and rejected in

the past. A simulation that indicates to a proposing party the likely supporting bloc and the likely opposing one would improve the chances of a *proposal* being adopted. A means of testing alternative wordings of the same *proposal* before submission would do the same.

Substantive aim To identify non-trivial patterns of interaction between the parties, and to predict the success of a *proposal* from an input text proposing a change to the *status quo*.

Methodological objective To use *stance* and blocking probability to build an agent-based simulation, and to construct a decision-support tool aggregating both measures.

Assumption The written record of the ATCMs is sufficient to model the consensus dynamics of interest. Geopolitical context is treated as constant within each meeting.

Research questions

- a. Can *stance* and blocking probability serve as inputs to an agent-based simulation that produces, for each *issue*, a ranking of potential coalition partners for a proposing party?
 - i. Which type of agent-based simulation is most suitable given the available input parameters?
 - ii. Which outputs are useful to a proposing party and derivable from the available measures?
 - iii. How accurately do the resulting rankings predict blocs of parties for out-of-sample *proposals* from recent ATCMs?

2.2 Resources and funding required

The project uses openly published data from the ATs document database, open-source and/or QUT-licensed software, and the QUT high-performance computing cluster. No additional funding is required.

2.3 Individual contribution to the research team

I am the sole researcher on this project, with guidance from my supervisory team. The Chapter 1 infrastructure developed at the outset of candidature is presented not as thesis results but as preliminary work establishing feasibility, and all reported findings will be produced within candidature. The thesis extends prior group work by Botelho et al. (2026), Kumar et al. (2026) and Carter et al. (2026).

2.4 Acknowledging use of Artificial Intelligence (AI) and AI-assisted technologies

I have used Claude Code for boilerplate and scaffolding code in the implementation of the pipelines and scripts. All architectural decisions, methodological choices, and algorithm design were specified by me in the input prompts. I also reviewed and edited generated code. I have also used Consensus AI and the Claude app to fetch the most relevant scientific papers from the available literature, based on my own ideas, keywords, and criteria. Afterwards, I read the papers and drew conclusions myself. Finally, I used Claude Code as a writing assistant for this document: rewording and help with grammar, readability, clarity and LaTeX document formatting. All research questions, project design, ideas, interpretation of results, and selection of literature are my own.

2.5 Timeline for completion

The candidature is full-time and commenced on the 31st of March, 2026 (see Table 2.1).

Table 2.1: Candidature timeline (April 2026–April 2029). ■ Ch. 1 ■ Ch. 2 ■ Ch. 3 ■ Milestones/events.



Table 2.1 (continued): Conference Attendance and Coursework.

Activity	3	6	9	12	15	18	21	24	27	30	33	36
	Jul	Oct	Jan	Apr	Jul	Oct	Jan	Apr	Jul	Oct	Jan	Apr
	'26	'26	'27	'27	'27	'27	'28	'28	'28	'28	'29	'29
Conference Attendance												
QANZIAM 2026												
AMEG Symposium 2026												
QANZIAM 2027												
International Conference												
AMEG Symposium 2027												
QANZIAM 2028												
Coursework												
RIO (completed)												
AIRS / IFN006												

Coursework (IFN006 Advanced Information Research Skills) is currently underway and will be completed by July 2026 following Stage 2 submission. The Research Integrity Online quiz is completed (see [Appendix A](#)).

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Appendices

Appendix A: Research Integrity Online

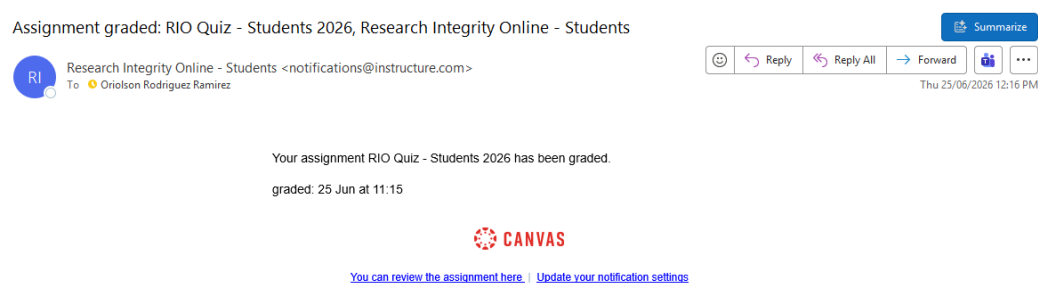


Figure 1: RIO completion notification email.

Grades for Oriolson Rodriguez Ramirez

Print Grades

Course: Research Integrity Online - ▾

Arrange by: Due date ▾

Apply

Name	Due	Submitted	Status	Score
RIO Quiz - Students 2026 RIO Quiz		25 Jun at 11:15		100 / 100
Assignments				
RIO Quiz				

Figure 2: RIO quiz completion record.

Appendix B: Coursework

AIRS / IFN006 is currently underway. The course ends in July 2026.

Appendix C: Notation

In the appendices, *topic* and *issue* are used interchangeably; the index k ranges over them.

	c, i	countries (Consultative Parties), $c \in \{1, \dots, C\}$
	k	topic, $k \in K$; $ K $ is the number of topics
	y	year
	d	document (Working Paper)
	p	passage (chunk) of a document
Indices and sets.	ℓ	ordinal stance class, $\ell \in \{1, 2, 3\}$
	$D(c, y)$	Working Papers co-authored by country c in year y
	$K(d)$	topics assigned to document d
	P_d	passages of document d , $P_d = \{p_1, \dots, p_{n_d}\}$
	$P_{d,k}$	passages of d selected as relevant to topic k
	A_d	co-authors of document d
	m	embedding dimensionality
	$\mathbf{e}_p$	embedding of passage p , $\mathbf{e}_p \in \mathbb{R}^m$
Embedding space.	$\boldsymbol{\mu}_k$	centroid of topic k , $\boldsymbol{\mu}_k \in \mathbb{R}^m$
	$\rho_{p,k}$	passage–topic relevance, $\rho_{p,k} = \cos(\mathbf{e}_p, \boldsymbol{\mu}_k)$
	τ	relevance selection threshold
	h_k^+, h_k^-	for- and against-pole hypotheses for topic k
	$\lambda(d, h)$	null-premise-debiased entailment log-ratio
Stance and hypotheses.	$\pi_{d,k}$	three-class ordinal distribution, $\pi_{d,k} \in \Delta_3$
	Δ_3	probability simplex, $\{\pi \in \mathbb{R}^3 : \pi_\ell \geq 0, \sum_\ell \pi_\ell = 1\}$
	v	class values, $v = (-1, 0, 1)$
	$s_{d,k}$	signed stance of d on topic k , $s_{d,k} \in [-1, 1]$

	$w_{d,c}$	attribution weight: 1 (full credit) or $1/ A_d $ (fractional)
	$S_{c,k,y}$	support of country c on topic k in year y , Eq. (3)
Aggregation and tensors.	$S_{\cdot,\cdot,y}$	year- y stance slice, $S_{\cdot,\cdot,y} \in \mathbb{R}^{C \times K }$
	$O_{c,k,y}$	opposition of country c on topic k in year y , $O_{c,k,y} \in [0, 1]$

For these tensors, high sparsity is expected because absence of evidence is recorded as missing, not as zero.

Appendix D: Opposition-layer workflow

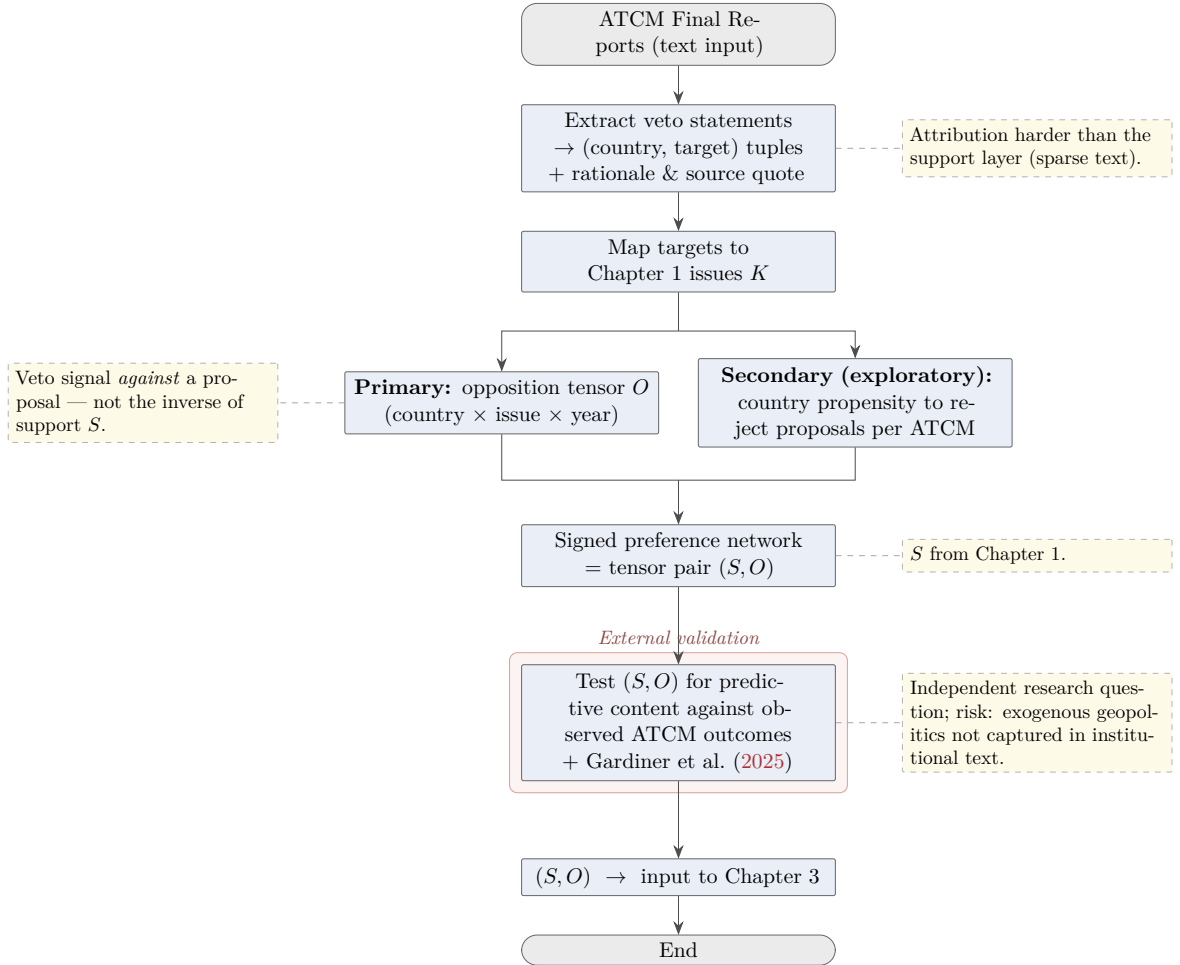


Figure 3: Chapter 2 opposition layer sketch: from text data to opposition tensor O , forming the signed preference network (S, O) .

Appendix E: Extended literature and method notes

The gap, literature by literature. The descriptive work of Section 1.4 (Gardiner et al. 2025; Mancilla and Jabour 2023) shows what the regime outputs but not who

produces what. The structural analyses of co-authorship networks (Botelho et al. 2026; Kumar et al. 2026) are undirected and unattributed. The text-based positional methods of Section 1.4 recover latent positions rather than topic-specific stances over time (Appendix G). Finally, consensus has been simulated as convergence (Section 1.2) or on synthetic data (Section 1.3).

ATCM document and output types. The ATCM’s inputs and outputs, including the paper types beyond Working Papers, are described in Appendix F.

Responsive (opinion-dynamics) model variants. Examples are the Kuramoto and opinion-changing-rate models used by Botelho et al. (2026), synchronization models that represent agreement as a coupled-oscillator convergence, and the bounded-confidence (Hegselmann and Krause 2002) and stubborn-agent (Friedkin and Johnsen 1990) variants of the iterative-averaging scheme of DeGroot (1974).

Veto-players condition (Tsebelis 2002). Let q denote the status quo, p a proposal, and u_i the utility function of party i , which scores how much that party values each policy. The status quo is replaced only when $u_i(p) \geq u_i(q)$ for every party i : no party may be left worse off than it is now.

Foundational MARL literature. Foundational MARL work covers multi-issue bargaining (Chang 2021), negotiating-team formation (Bachrach et al. 2020), bargaining under risk and asymmetric information (Schmid et al. 2020), and search-improved game-theoretic learning (Li et al. 2023). In these systems, neural networks (NN) enter only as function approximators, allowing the learning to scale to large state and action spaces.

LLM-based negotiation simulations. Moghimifar et al. (2024) simulate coalition formation in parliament using text data from party manifestos. Bolleddu (2025) present a MARL system for consensus building and negotiation based on synthetic data. Kulkarni et al. (2025) use negotiated agreements with game-theoretic reasoning to predict dynamic coalition formation in Diplomacy, a board game in which seven powers negotiate, form, and break alliances, widely used as a benchmark for this type of simulation. Some benchmarks for real negotiation data are starting to appear (Benac et al. 2026).

Wordfish (Slapin and Proksch 2008). Its aim was to estimate party positions over time without hand-coded reference texts. The model treats each word count as

a Poisson draw whose expected frequency depends on the party’s position on a single latent dimension, and it recovered credible left–right trajectories for German parties from their election manifestos (1990–2005).

Wordshoal (Lauderdale and Herzog 2016). Their model scales each debate separately with Wordfish and then combines the debate-level positions into a general dimension through a Bayesian factor analysis. Applied to speeches in the Irish Dáil and the US Senate, it recovered within-party disagreement that voting records hide. Each debate scale is whatever axis dominates that debate’s word frequencies, which need not be the substantive for-or-against axis of interest. Lauderdale and Herzog get away with this because legislative debates are adversarial, so the dominant axis is usually government versus opposition; ATCM Working Papers have no such structure.

BERTopic and NLI models. BERTopic (Grootendorst 2022) clusters document embeddings and labels each cluster with its most distinctive words. Natural language inference (NLI) models (Burnham 2023; Abercrombie and Batista-Navarro 2022) are classifiers that take a text and a stated hypothesis (e.g. “this paper supports stricter protection”) and judge whether the text entails, contradicts, or is neutral towards it. A single stable ideal point is too coarse for the ATCM, where a party can support stricter environmental protection yet oppose new tourism rules.

Support-layer pipeline details. Each document is embedded: mapped to a vector in $\mathbb{R}^m$, under the geometric assumption that documents with similar meaning receive nearby vectors. BERTopic reduces the dimension of these vectors and clusters them, and each cluster is a topic; each centroid $\boldsymbol{\mu}_k \in \mathbb{R}^m$ is the average embedding of the documents in topic k . Each document d is split into passages $P_d = \{p_1, \dots, p_{n_d}\}$, and each passage p is embedded as $\mathbf{e}_p \in \mathbb{R}^m$. Relevance is defined quantitatively: $\rho_{p,k} = \cos(\mathbf{e}_p, \boldsymbol{\mu}_k)$, the cosine similarity between passage embedding and topic centroid, so a passage that points in the same direction as the topic centroid discusses that topic. Relevance acts as a gate: $P_{d,k}$ is the set of passages scoring above a threshold, and only these pass to the next stage. The topic-relevant sentences in each $P_{d,k}$ are extracted using a Large Language Model (LLM); $\text{Extract}(P_{d,k})$ denotes this step. The extracted sentences are concatenated and scored as a set by the NLI classifier: given the text as premise and a hypothesis about topic k , it scores entailment, neutrality, and contradiction. Entailment means the premise gives support to the hypothesis. Contradiction means the premise contradicts it. The resulting scores are combined into three ordinal classes,

$$\pi_{d,k} = \text{NLI}(\text{Extract}(P_{d,k})) \in \Delta_3, \quad \Delta_3 = \left\{ \pi \in \mathbb{R}^3 \mid \pi_\ell \geq 0, \sum_{\ell=1}^3 \pi_\ell = 1 \right\}, \quad (1)$$

where Δ_3 is the probability simplex: $\pi_{d,k}$ assigns a non-negative probability to each of the three classes, against, neutral and for, and these probabilities sum to one. The signed stance is the expected value of this distribution under an equal-spacing convention,

$$s_{d,k} = \sum_{\ell=1}^3 v_\ell \pi_{d,k}(\ell), \quad v = (-1, 0, 1). \quad (2)$$

Since $\pi_{d,k}$ is a probability distribution and the three fixed class values v_ℓ all lie in $[-1, 1]$, the stance $s_{d,k}$ is a convex combination of those values. The stance is therefore a continuous quantity in $[-1, 1]$ on topic k ’s axis, while v itself never varies.

Recovering stance as a contrast (Chapter 1). Measuring directional stance in diplomatic text is challenging because diplomats hedge: cautious, non-committal phrasing and a shared vocabulary of safeguards pull single-hypothesis scores toward neutrality. A party rarely writes “we oppose this measure”, it writes “further consideration may be warranted”, and a phrase such as “subject to appropriate environmental safeguards” appears in permissive and restrictive papers alike. Stance is therefore recovered as a contrast. For each issue, two author-anchored hypotheses are written: a supporting pole and an opposing pole (e.g. h_k^+ “the author of this paper supports further regulation of tourism”, h_k^- “the author of this paper supports relaxing tourism regulation”). Each is paired with the extracted text as premise and scored by the NLI classifier (Laurer et al. 2024), and the against-pole score is subtracted from the for-pole score, giving a direction that survives the non-committal phrasing. As a robustness check, the same texts are scored a second time by a generative LLM used as a secondary instrument (Le Mens and Gallego 2025).

Stance classifier (Chapter 1). For topic k , let h_k^+ and h_k^- be the for- and against-pole hypotheses. For passage set d and hypothesis h , the debiased evidence is the null-premise-corrected log-ratio,

$$\lambda(d, h) = \text{clip}\left(\log P(\text{entail} \mid d, h) - \log P(\text{entail} \mid \emptyset, h), \pm 5\right),$$

this removes each hypothesis’s marginal acceptability. The three-class distribution combines entailment of each pole with contradiction of its opposite, and takes the neutral class from the classifier’s own neutrality scores,

$$\pi_{d,k} = \text{softmax} \begin{pmatrix} \lambda_{\text{ent}}(h_k^+) + \lambda_{\text{con}}(h_k^-) \\ \lambda_{\text{neu}}(h_k^+) + \lambda_{\text{neu}}(h_k^-) \\ \lambda_{\text{ent}}(h_k^-) + \lambda_{\text{con}}(h_k^+) \end{pmatrix},$$

where the rows are the for, neutral and against classes. A paper that hedges scores neutral against both poles, so it falls in the middle class rather than at either end. The scalar stance is then the expected value of Equation 2.

Aggregation (Chapter 1). The document stances are aggregated over the Working Papers, weighted by co-authorship credit,

$$S_{c,k,y} = \sum_{\substack{d \in D(c,y) \\ k \in K(d)}} w_{d,c} s_{d,k}, \quad (3)$$

where $S_{c,k,y}$ is the support of country c on topic k in year y , $D(c,y)$ is the set of Working Papers co-authored by country c in year y , $K(d)$ is the set of topics assigned to document d , and $w_{d,c}$ is the attribution weight of that country in one document. For a fixed year y , the slice $S_{\cdot,\cdot,y} \in \mathbb{R}^{C \times |K|}$ is a signed country $\times$ topic matrix, with C countries and $|K|$ topics; a dot means the index runs over all its values.

Attribution weights (Chapter 1). Two attribution conventions are computed:

- Full credit, $w_{d,c} = 1$: every co-author fully holds the position. This is the coalition-membership reading (“who is on this side?”).
- Fractional credit, $w_{d,c} = 1/|A_d|$, with A_d the co-authors of d : the paper’s positional mass is divided among its authors, discounting states that merely sign on. This is the influence reading (“who is driving this?”).

Other ideal-point precedents. Vafa et al. (2020) produced a probabilistic framework that jointly places ideal points and models actors’ language.

Embeddings. An embedding may cover a word, a sentence, or a whole document. The map is learned from large corpora so that texts appearing in similar contexts receive nearby vectors. Neural language models such as BERT (Devlin et al. 2019) make the vectors context-dependent: the same word receives different vectors in different sentences.

The UN General Debate corpus (Baturu et al. 2017). Their corpus collects over 7300 annual General Debate statements (1970–2014), one per country per year, in a setting that parallels the ATCM: formal, annual, state-level, and translated into official languages. They derive preferences in two ways, both from word frequencies: Wordscores, a supervised scaling in which the analyst defines the axis in advance through reference

texts (US and Russian statements anchor a US–Russia dimension), and correspondence analysis, an unsupervised reduction whose dimensions are interpreted afterwards.

Opposition tensor (Chapter 2). $O_{c,k,y} \in [0, 1]$ is the opposition of country c on topic k in year y , built from the extracted (*country*, *target*) pairs: 0 means the country is not likely at all to veto, and 1 means a certain veto.

Simulation-layer equations (Chapter 3). Each agent’s action on a presented proposal is binary, $o_i \in \{0, 1\}$: one if agent i objects, zero otherwise. Under the consensus rule the proposal is adopted only if no Party objects,

$$1_{\text{adopt}} = \prod_j (1 - o_j), \quad (4)$$

therefore, a single objection vetoes adoption. The reward to agent i is then

$$R_i = \alpha S_{i,k,y} 1_{\text{adopt}} - \beta O_{i,k,y} 1_{\text{adopt}} - \gamma o_i, \quad (5)$$

where the first term rewards adoption of outcomes the Party supports, the second penalises adoption of outcomes it opposes, the third subtracts a diplomatic cost for casting an objection, and $\alpha, \beta, \gamma \geq 0$ are trade-off weights, learned during training to fit the historical consensus record.

Appendix F: The ATs as a consensus regime

At the ATCM, Consultative Parties (CPs) reach decisions exclusively by consensus; non-consultative parties (NCPs), observers, and experts attend but take no part. These outputs take three forms (Secretariat of the Antarctic Treaty 2023, Rule 24). **Measures** set out substantive provisions that the parties intend to be legally binding in their domestic law; they require each party’s government to formally approve them. **Decisions** settle the ATCM’s own internal and organisational matters. **Resolutions** are non-binding: they record recommendations or expressions of intent, with no legal force.

There are four types of input. The one that matters for this thesis is the **Working Paper (WP)** (Secretariat of the Antarctic Treaty 2023, Rule 48): submitted by CPs and Observers, WPs require discussion at the ATCM and are the main source of outputs. They are the only input from CPs that can produce an outcome. **Secretariat Papers** (Secretariat of the Antarctic Treaty 2023, Rule 49) cover organisational matters such as the budget. **Information Papers** (Secretariat of the Antarctic Treaty 2023, Rule 50)

provide supporting information, also from NCPs and experts. **Background Papers** (Secretariat of the Antarctic Treaty 2023, Rule 51) are documents submitted for the record only.

The consensus system makes the ATs a notable example of international cooperation (Jackson 2021); however, the same mechanism has become a major challenge as the number of CPs grows (Tsebelis 2002; Gardiner et al. 2025).

A notable case of contested consensus occurred in the late 1970s and 1980s, during the negotiations over mineral resources in Antarctica. The Convention on the Regulation of Antarctic Mineral Resource Activities (CRAMRA) was a framework, agreed in 1988, that would have permitted and regulated mining in Antarctica. It was abandoned before entering into force and replaced by the Madrid Protocol (1991), which instead prohibited all mineral resource extraction and designated Antarctica a natural reserve devoted to peace and science (Jackson 2021). Although it unfolded in two stages, it forms a single documented episode: one negotiation over Antarctic mineral resources in which the parties' positions are recorded and then reversed. That documented reversal makes it a valuable canary test, a known case against which any quantitative stance measure can be validated.

Appendix G: Measuring policy preference from political text

Positional measures place each actor on a policy scale, for example from favouring regulation to favouring access, using only the text the actor produces. Extracting them from political texts is a well-developed field with two paradigms. They differ in how they turn text into numbers: the first paradigm uses word counts (i.e. frequencies), while the second uses embeddings. The first paradigm is scaling and ideal-point methods, which recover a latent positional dimension from the data itself: the axis is not chosen in advance, it emerges from patterns in the text, and its meaning is interpreted after estimation. Wordfish (Slapin and Proksch 2008) is a Poisson scaling model of party text (Appendix E). Its limitation is that the recovered dimension is uninterpreted: it is the dominant axis of variation in word usage, and in a corpus as topically diverse as the ATCM Working Papers it would likely reflect what parties talk about rather than what they support. Lauderdale and Herzog (2016) addressed the problem that legislators talk about different things in different debates by scaling each debate separately, but the debate scales inherit the same uninterpreted-dimension problem, and the debates must be given in advance (Appendix E). Two things are therefore missing for this thesis: the topics have to be discovered from the papers themselves, and the output must

be a stance per topic rather than one general dimension. The closest dataset to the ATCM corpus is the UN debate corpus Baturo et al. (2017) use to derive state-level preferences, in a setting that parallels the ATCM (Appendix E). Finally, Parschan and Jakob (2025) review the text-based ideal-point estimation algorithms of this first paradigm, comparing their assumptions and variants in one place. The second paradigm uses embeddings as its main tool. An embedding maps a piece of text to a vector in $\mathbb{R}^d$ so that texts with similar meaning receive nearby vectors: comparisons of meaning become geometric operations. Modern embeddings are produced by neural language models such as BERT (Devlin et al. 2019) (details in Appendix E). Word-embedding geometry recovers ideological placement in parliamentary data (Rheault and Cochrane 2019), but the choice of embeddings strongly affects applied inference and the results must be validated (Rodriguez and Spirling 2022).

This thesis adopts the second paradigm and uses embeddings at several stages: to discover what topics the Working Papers discuss, to measure how relevant each paper is to each topic, and, through the same family of neural models, to read each paper’s stance on a topic. This yields the topic-conditioned stance that the first paradigm cannot supply: a separate position per party and topic, allowed to change through time. The topic structure is discovered using BERTopic (Grootendorst 2022) and the stance is read with natural language inference (NLI) models (Burnham 2023; Abercrombie and Batista-Navarro 2022), both described in Appendix E. Section 2.1.2 sets out the validation checks for these results: against the Secretariat’s hand-assigned topic categories, against the documented CRAMRA reversal, and against observed co-authorship.

Appendix H: Preliminary results of Chapter 1

During these months of work, some progress has been made in implementing the BERTopic (Grootendorst 2022) pipeline and two extra stages for relevance selection and stance. These are preliminary results that could change. The following figures show that BERTopic is effectively capturing some of the known issues that have been discussed at the ATCM, for example the mineral dispute. It also shows clear clusters, for example historic hut heritage and disease in native species. There is still some noise that needs more refinement. The stance stage is already capturing some of the countries’ positions over the years, though it still shows too many neutral stances in the oldest years.

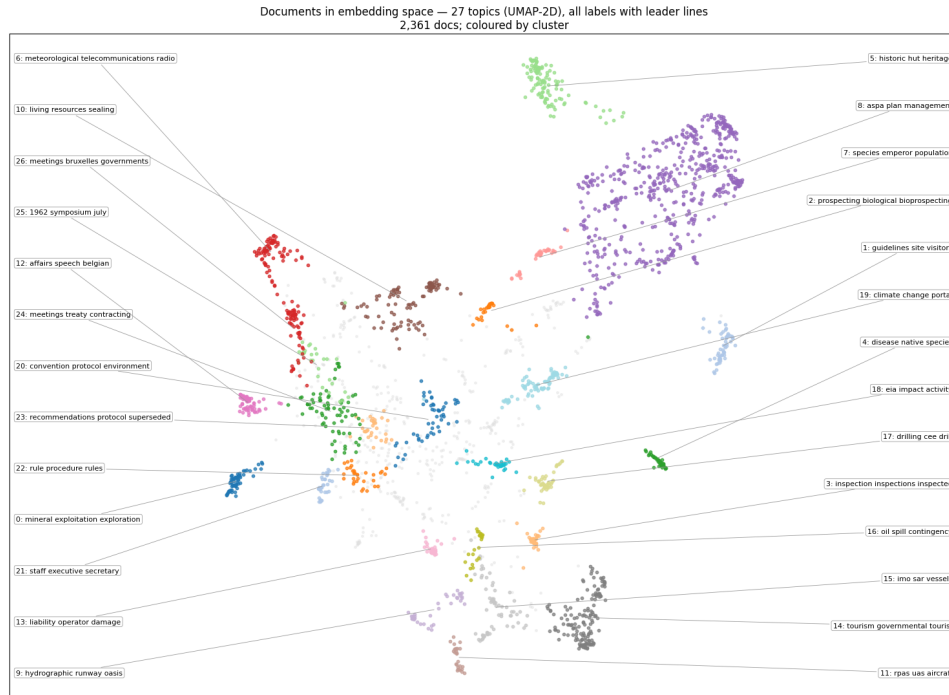


Figure 4: Topic landscape of the Working Papers. Each point is a document; colours are the BERTopic clusters and grey points are unclustered (HDBSCAN noise). Recognisable clusters include the mineral dispute, historic hut heritage, and disease in native species.

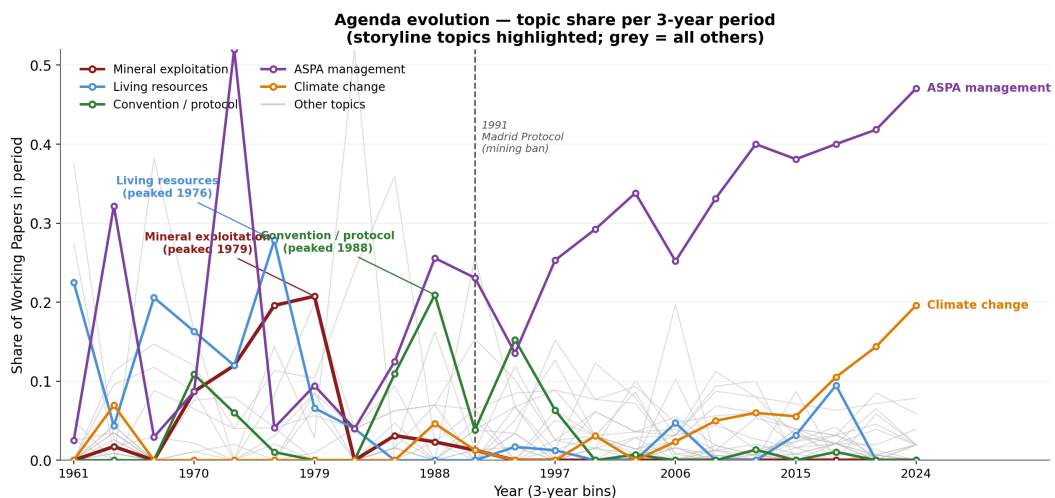


Figure 5: Topic prevalence over time (three-year bins). The pipeline captures the mineral episode, which peaks in the 1970s–80s and declines after the 1991 Madrid Protocol, as well as recent trends such as Antarctic Specially Protected Areas (ASPA) and climate change.

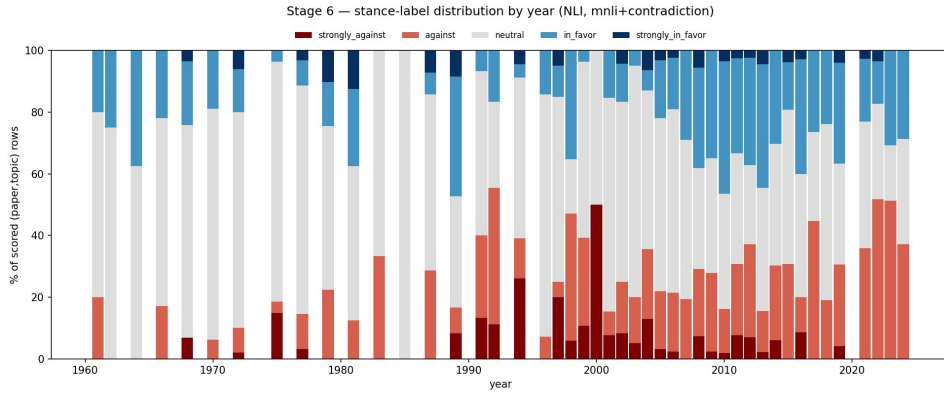


Figure 6: Recovered stance over time across the scored (paper, topic) rows. The neutral band shrinks from the 1960s–80s to the 2010s–20s; the oldest years still show many neutral stances and rest on few papers.